C. Minimising the distance

We search over candidate parameter vectors $\theta = (\text{mean}, \ \kappa)$ of the Beta growth regime, computing the Wasserstein-1 distance

$$W_1(F_{\text{obs}}, \ F_{\boldsymbol{\theta}})$$

at each grid point. The heatmap shows $\log_{10}$ distance: dark regions are far from the observed CDF; light regions are close.

A two-stage procedure—coarse grid scan followed by L-BFGS-B refinement—locates $\hat{\theta}$, the parameter vector that minimises the distance between the predicted and observed marginal CDFs of V.